

**Enhancing the Gross Motor Skills of Children with
Down Syndrome**

24-25J-096

FINAL REPORT

Wijesinghe S.A.A.K

(IT21338120)

B.Sc. (Hons) In Information Technology, Specialized in Software
Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2025

Declaration

“I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Signature:

Date:

The supervisor/s should certify the dissertation with the following declaration. The above candidate has carried out research for the bachelor’s degree Dissertation under my supervision.

Signature of the supervisor:

Date

Abstract

Down syndrome (DS) is a genetic disorder affecting cognitive and motor development, which results in delays in physical activity and coordination. Early intervention through the use of special physical exercises can significantly improve the motor function and life quality of Down syndrome [1].

Despite all these advances, there is still a need for more individual and automated procedures for assessment and development of motor skills in DS children. Recent studies have emphasized the need for promoting physical activity [2] and have highlighted issues of information processing and movement control [3] in DS children. To address the above issues, advanced computational techniques, in the form of deep learning models, have been largely applied in the area of action recognition, taking advantage of skeleton information for detecting and understanding motor actions [4]. In the course of this research, the assessment and enhancement of the motor abilities among Down syndrome will be enhanced by employing machine learning techniques to skeleton-based action recognition. By analyzing key points of motion data, this technique permits automatic movement pattern detection, producing a more accurate motor skill level determination. The Data set is formed of skeletal key points from video information, providing for a sophisticated, non-invasive way to monitor motor skills chronologically.

Key words: Down Syndrome, Cognitive and motor development, Physical Activity, Skeleton based , Machine Learning, Action Recognition

Acknowledgement

I would like to express my heartfelt gratitude to all those who assisted me during the successful completion of this research.

Firstly, I am sincerely grateful to my supervisor, Ms. Jenny Krishara.

Her valuable advice, gentle guidance, and constant encouragement were a source of strength and directed this research in the right direction.

I also would like to convey my deepest gratitude towards my co-supervisor, Ms. Tamali Kelagama. Her technical expertise, keen criticisms, and guidance in shaping my concepts played a critical role in advancing the quality of this work.

My heartfelt appreciation is also given to the Department of Information Technology, Sri Lanka Institute of Information Technology (SLIIT). Their advanced facilities, research equipment, and inspiring learning environment played a vital role in the success of the project.

A special appreciation is owed to my external supervisor, Dr. Sudarshi Lenagala, who gives me knowledge about how to deal with children with Down Syndrome, what their limitations are. And she shared his own experiences throughout the research as a doctor.

I would also like to thank my dedicated team members. Their continuous support, hard work, and team spirit enabled me to conquer every hurdle successfully. Their innovative thoughts and technical expertise were vital in achieving the goals of this research.

Finally, I wish to thank my own dear mother. Her endless love, understanding, and unwavering support were my own source of courage. Her blessings and belief in me proved vital to overcoming the difficulties experienced on this journey.

Without the strong encouragement, support, and guidance given by all these great people, it would have been impossible to successfully complete this research. I take my deepest and sincerest gratitude to them all.

Table of Contents

Declaration	i
Abstract	ii
Acknowledgement	iii
List of figures	vii
List of Tables.....	viii
List of abbreviations	ix
1. Introduction.....	1
1.1. Background and Literature Review	3
1.2. Research Gap	5

1.3.	Research Problem	6
	Research Objectives.....	7
	Main Objective.....	7
	Specific Objectives	7
2.	Methodology	9
2.1.	Methodology	9
2.1.1.	Gross motor skills selection	9
2.1.2.	Project Requirements	12
2.1.2.1.	Functional Requirements	12
2.1.2.2.	Non-functional	12
2.1.2.3.	Software Requirements	12
2.1.3.	Project Breakdown.....	13
2.1.3.1.	Grant Chart.....	13
2.1.3.2.	Work Breakdown Chart	13
2.1.4.	Design Diagram	14
2.1.4.1.	Use Case Diagram.....	14
2.1.4.2.	Sequence Diagram	Error! Bookmark not defined.
2.1.5.	Research Domain	15
2.2	Commercialization	16
2.3	Testing and Implementation.....	17
2.3.1	Collecting video inputs for dataset	17
2.3.2	Data Preprocessing.....	19
2.3.3	Train Model.....	23
2.3.4	Developing gross motor skills research component process	27
2.3.5	Categorize into levels.....	30
2.3.6	Provide gross motor skills based on the selected level	35
3.	Results & Discussions.....	44
3.1	Result	44
3.2.....		45
3.3.....		46
4.	Conclusion	47

5. References	48
---------------------	----

List of figures

Figure 1: External Supervisor approval for Gross motor skills	11
Figure 2: Grant Chart	13
Figure 3: Work Breakdown Chart	13
Figure 4: Use Case Digram	14
Figure 5: Libraries in used in Collect.py	17
Figure 6: MediaPipe	18
Figure 7: Extract Key points	18
Figure 8: Convert BRG into RGB	19
Figure 9: Same man with Black background	19
Figure 10: Man with Skeleton	19
Figure 11: Find the skeleton	20
Figure 12: Reshape the dataset	20
Figure 13: Save file with .npy	21
Figure 14: Save the dataset	21
Figure 15: Collected .npy dataset	22
Figure 16: Libraries used in training	23
Figure 17: Label the classes	23
Figure 18: Train set and Test set	24
Figure 19: Training Dataset	24
Figure 20: Use 50 epochs	25
Figure 21: Loss function, Data Loader	26
Figure 22: Model accuracy and loss	27
Figure 23: Frontend Interface	28
Figure 24: When click on the gross motor skill	28
Figure 25: Generate random video	29
Figure 26: Generate Random vidoe	30
Figure 27: Level categorization	30
Figure 28: Extract Key points	31
Figure 29: Level Selection Percentage	32
Figure 30: Output	32
Figure 31: Child select to Easy Level	33
Figure 32: Child Select to Hard Level	33
Figure 33: Child select to Medium Level	34
Figure 34: No human detected	34

List of Tables

Table 1: Test Case – GMS01	35
Table 2: Test Case - GMS02	36
Table 3: Test Case – GMS03	36
Table 4: Test Case – GMS04	37
Table 5: Test Case – GMS05	37
Table 6: Test Case – GMS06	37
Table 7: Test Case – GMS07	38
Table 8: Test Case – GMS08	38
Table 9: Test Case – GMS09	38
Table 10: Test Case – GMS10	39
Table 11: Test Case – GMS11	39
Table 12: Test Case – GMS12	40
Table 13: Test Case – GMS13	40
Table 14: Test Case – GMS14	40
Table 15: Test Case – GMS15	41
Table 16: Test Case – GMS16	41
Table 17: Test Case – GMS17	42
Table 18: Test Case – GMS18	42
Table 19: Test Case – GMS19	42
Table 20: Test Case – GMS20	43
Table 21: Test Case – GMS21	43

List of abbreviations

Abbreviation	Description
DS	Down Syndrome
LSTM	Long
RNN	Recurrent Neural Network
PA	Physical Activity

1. Introduction

Down Syndrome occurs in the presence of an extra chromosome. This will affect approximately 1 in 700 newborns. Often children with Down Syndrome have short arms and legs compared to their body. DS children have low muscle tone (hypotonia) and loose-jointedness (hyper flexibility), which are important for engaging in gross motor skills. The growth of their bodies is usually slower than other children [1].

Children with Down Syndrome are not active in physical activities. PA means any movements of the skeletal muscles that throw out energy. Because of the lack of doing PA, children with DS can be victims of obesity, gait disorders, hip dysplasia and many more non-spread diseases. Physical therapists mostly focus on balancing DS children. DS children are also pushed into doing group activities and games to improve social behavior [2]. Goal-directed movements focus on how individuals perform the given actions in different environments. Processing this information helps learning and improving future movements [3]. Children with Down Syndrome often face unique challenges in their physical development, particularly in acquiring gross motor skills, developing gross motor skills, vital for their overall physical development and independence. These skills involve large muscle movements such as walking, running, balancing and are fundamental to a child's overall physical health, independence and ability to engage in social activities. Developing gross motor skills not only contributes to physical fitness but also plays a significant role in cognitive and social development of gross motor skills crucial for these children as it lays the foundation for more complex physical activities and contributes to their cognitive and social development by enabling children to explore their environment, interact with peers and participate in group activities. Early and targeted interventions focused on gross motor enhancement are essential, as they lay the groundwork for more complex activities and contribute significantly to a child's overall fitness and quality of life.

However, due to the low muscle tone (hypotonia), joint laxity of children with Down Syndrome may experience delays in engaging in gross motor activities. Traditional methods of teaching gross motor skills often rely on repetitive exercises and physical therapy sessions, which may not sufficiently engage or be adaptable to each child's individual needs.

In response to these challenges, there is a growing interest in leveraging technology to create more engaging and personalized learning experiences. This project aims to develop a system that enhances the gross motor skills of DS children through the use of supporting video clips and task recognition models. By providing continuous and adaptive learning opportunities, the system allows children to progress at their own pace and receive immediate feedback, making the learning process both effective and

enjoyable. This approach seeks to promote the physical development and overall well-being of children with Down Syndrome by categorizing them into levels and giving them gross motor activities based on that.

This research focuses on developing the gross motor skills of children with Down Syndrome and improving them. Use skeleton to perfectly identify the action of the child doing. The importance of using skeleton is it enables accurate action recognition even in challenging scenarios such as complex backgrounds, poor lighting situations and different environments [4]

1.1. Background and Literature Review

In order to improve the general physical development and independence of children with Down syndrome (DS), this research project focuses on improving their gross motor skills. The development of these motor skills is greatly aided by physical activities (PA). Children who participate in different types of physical activity not only develop their motor skills but also feel happier, more motivated to play sports, and have healthier bodies in general. It has been demonstrated that these activities help children with Down syndrome by fostering their physical strength, confidence, and coordination—all of which enhance their overall well-being and personal growth. Gross motor skills aid in a child's physical development by helping them coordinate movements like standing, walking, running, and jumping.

The two main categories of motor skills are fine motor skills and gross motor skills. Standing, walking, running, and jumping are examples of gross motor skills, which are the abilities needed to make large movements with the body's limbs. Fine motor skills, on the other hand, include more precise, smaller movements, like writing or grasping objects with the fingers. A child's overall development and well-being depends on the development of these motor skills, with gross motor skills being especially important for laying the groundwork for more complex movements.

The ability to use these motor skills effectively is directly related to physical activity. It is especially crucial for children's development of the motor skills they will need as they grow up. Early motor skill development has long-term advantages that affect not only physical health but also social and cognitive development, according to research. Physical activity becomes even more important for the general growth and health of children with Down syndrome, who are generally delayed in developing gross motor skills [5]. Delays in the development of motor skills can lead to several health issues, such as a higher chance of non-communicable diseases and ailments linked to a lack of physical fitness [2].

Delays in the development of motor skills in children with Down syndrome can also significantly affect their mental and cognitive growth. Since gross motor skills require coordination as well as an understanding of movement and spatial awareness, they are closely linked to a child's mental development. Since more complex tasks require greater speed, coordination, and muscle control, the development of motor skills becomes more crucial for kids over the age of six [6]. These skills, whose development is frequently delayed in children with Down syndrome, are essential for the child's independence and involvement in everyday activities.

A child's physical development depends on their ability to coordinate fundamental movements like standing, walking, running, and jumping. However, the combination of physical and cognitive factors frequently causes a delay in reaching these motor milestones in children with Down syndrome. Hypotonia, a disorder marked by low

muscle tone, is one of the main issues that children with Down syndrome face. Children struggle to complete tasks requiring muscle control, balance, and stability as a result of their diminished muscle strength and endurance. Because of this, these kids might have a hard time doing even simple physical tasks.

The delayed development of gross motor skills in children with Down syndrome is largely caused by balance problems in addition to muscle control problems. These kids may find it much more difficult to run, jump, and walk efficiently due to a lack of balance and coordination. Even basic physical tasks become more challenging for kids with these issues, which causes delays in the development of fundamental motor skills. Their capacity to carry out independent tasks may suffer as a result, becoming more dependent on outside help for tasks that most kids could normally complete on their own.

Children with Down syndrome may experience social isolation and low self-esteem as a result of developmental delays in gross motor skills. Less interaction with peers and fewer opportunities for social engagement may result from children who struggle with motor skills finding it more difficult to participate in group activities. These kids might feel excluded as a result, which could further impair their social skills. Their sense of self-worth may change as a result of tasks that become more challenging or impossible to complete, and they may start to feel incapable or undeserving of reaching particular objectives.

A child's sense of autonomy may also be impacted by their incapacity to carry out tasks on their own. Being able to complete tasks independently, without help from others, promotes self-confidence and independence. The freedom to perform simple tasks enhances a child's emotional health in addition to their physical development. These kids will feel more empowered and accomplished when they have more chances to do motor activities on their own, which will motivate them to take part in more activities and continue their development.

Early intervention is essential to address these issues. As soon as possible, preferably in the early years of childhood, efforts should be made to mitigate the developmental delays in motor skills. It has been demonstrated that early participation in appropriate physical activities and physical therapy significantly improves a child's motor development, enabling them to become more independent and coordinated. Early intervention is essential for maximizing a child's potential and making sure they receive the help they need to develop motor skills, which will help them become more self-sufficient and have better general health throughout their lives.

1.2. Research Gap

Numerous studies have made a substantial contribution to helping kids with Down syndrome in a number of areas, including learning, evaluating motor skills, and identifying activities. However, when compared to the goal of this research, which is to improve gross motor skills in children with Down syndrome using AI-based solutions, these previous studies show a number of limitations

A teaching Method for Students with Down Syndrome is to support academic learning through personalized lesson plans. Although it offers useful teaching methods, it ignores the development of gross motor and physical skills, which are equally important for kids with Down syndrome. On the other hand, the suggested study offers a useful alternative to classroom instruction by focusing on improving gross motor skills directly through AI-driven action recognition and real-time feedback [7].

Fine Motor Tasks focuses on analyzing fine motor skills by examining muscle activity using sEMG signals. However, the implementation of this method necessitates specific equipment and focuses solely on fine motor skills, such as hand and finger movements. This is a major drawback for everyday use, particularly in domestic settings. By using basic video-based analysis to identify gross motor actions like walking, running, jumping, and throwing without the need for pricey or complicated hardware, the proposed study fills this gap [8].

In a similar vein, No Limit: A Down Syndrome Children's Educational Game places a strong emphasis on using gaming to develop cognitive skills. Although this research uses interactive games to improve intellectual abilities, it does not focus on improving physical motor skills. By emphasizing practical physical activities and offering tailored activity suggestions to enhance gross motor skills in kids with Down syndrome, the proposed study fills this gap. [9]

Additionally, research like The Effectiveness of Computer-Aided Video Modeling in Teaching Basic Basketball Movements to Individuals with Down Syndrome focuses on using video modeling techniques to teach basketball and other sports skills. Although useful, it is specific to a single sport and does not apply to a broader variety of fundamental motor skills. This study, on the other hand, looks at a wide variety of basic gross motor skills outside of sports-specific motions.[10]

Comprehensive Assessment and Early Prediction of Gross Motor Performance in Toddlers With Graph Convolutional Networks-Based Deep Learning is one study that focuses on early assessment using state-of-the-art AI techniques like Graph Convolutional Networks. However, these studies don't provide any strategies for skill improvement and mostly conclude with assessment. The proposed system gets

around this restriction by combining assessment and intervention through personalized activity recommendations and real-time feedback mechanisms. [11]

The methods used in other studies, such as Machine Learning Algorithms for Activity Recognition in Ambulant Children and Adolescents with Cerebral Palsy, are not directly applicable to children with Down syndrome because they concentrate on cerebral palsy; instead, they use machine learning models to target activity recognition for mobility assessment. The specific motor development requirements of children with Down syndrome are taken into consideration when designing the proposed research's solution.[12]

Furthermore, studies like the Multiview Child Motor Development Dataset for AI-driven Assessment of Child Development offer useful datasets for evaluation but lack tools for skill development or intervention. On the other hand, this study makes use of these evaluation results to inform tailored interventions meant to enhance gross motor skills.[13]

Similarly, the use of VR-based interventions in motor learning is highlighted in Motor Learning and Virtual Reality in Down Syndrome – A Literature Review. VR interventions, however, have accessibility problems for routine at-home practice and necessitate costly equipment. The suggested study uses common video input from gadgets like laptops and smartphones to provide an inexpensive, non-VR-based solution.[14]

Automatic Movement Recognition for Evaluating the Gross Motor Development of Infants uses machine learning and video-based movement analysis to detect motor abnormalities early. It does not, however, address older children or offer intervention strategies; instead, it focuses primarily on infants. The suggested system offers both assessment and intervention components for efficient gross motor skill development, and it targets children with Down syndrome across a wider age range.[15]

1.3. Research Problem

The development of gross motor skills, which are necessary for carrying out large-scale movements like walking, running, jumping, and balancing, is frequently significantly delayed in children with Down syndrome. The combination of cognitive and physical difficulties is primarily to blame for these delays. These kids might therefore find it difficult to meet developmental milestones at the same rate as their classmates who are normally developing. Their general physical health, coordination, and self-confidence may all be significantly impacted by this delay, which may then have an impact on their capacity to carry out daily tasks and communicate with others.

It is essential to use interactive teaching strategies that are tailored to the cognitive and learning preferences of children with Down syndrome in order to support their motor skill development. These kids can be used to creating an engaging learning environment because they are frequently more receptive to visual and auditory stimuli. The child can more easily comprehend what is expected of them if, for instance, interactive models, animations, or video demonstrations are used to clearly illustrate the desired actions. Furthermore, music or audio cues can enhance the learning experience and support the child's motivation and focus throughout the task.

1.4. Research Objectives

1.4.1. Main Objective

The main goal of this research is to create an interactive learning system that enhances the gross motor skills of kids with Down syndrome by using task and level recognition along with easily understood action video clips. The system will be built to offer performance evaluation and feedback in real time. Through a customized learning process that adapts to their needs and progress, this method not only improves their motor skills but also inspires them..

1.4.2. Specific Objectives

- **Identify the child level (Easy, Medium, Hard)**

Determining the child's present skill level is the first stage in developing a customized motor skills development plan. In order to do this, the system will:

- Using video analysis, evaluate the child's proficiency in a basic gross motor skill (such as walking, standing, or catching).
- A task difficulty classification system will be used to examine the child's movements, where:
 - A task will be classified as Easy Level if it is too difficult for the child to complete successfully.
 - The task will be classified as Medium Level if the child completes it successfully despite having some difficulty.
 - The task will be classified as Hard Level if the child can complete it with assurance and using the right technique.
- Based on this classification, the system will then give the child a particular level. The level aids in figuring out how challenging upcoming activities and assignments that are catered to the child's present skill level will be.
- By using this classification system, the exercises are kept from being too easy or too hard, which could demotivate the child. Rather, the child will be motivated to work on assignments that correspond to their level of proficiency.

- **Interactive video clips to perfectly learn the given gross motor activity**

Interactive video clips that include the following components will be used to make the learning process interesting:

- The gross motor skill will be demonstrated in clear video by the system. These videos will demonstrate the movement in real time and include clear, step-by-step instructions.
- To assist the child in comprehending and practicing each skill, the system will incorporate interactive features like animations, visual cues, and feedback mechanisms.
- The system will incorporate lively and entertaining background music that goes well with the educational video because kids with Down syndrome frequently react favorably to music. The child will be inspired by this musical component, which will also help them stay focused and pay attention..
- Since children with special needs frequently benefit from multisensory learning methods, the use of videos makes the learning process both visual and auditory.

- **Based on the child's level, provide personalized gross motor skill activities**

The system will adjust to offer tailored activities after determining the child's skill level:

- Activities suitable for the child's categorized level (Easy, Medium, Hard) will be suggested by the system.
 - Simpler exercises like walking, standing and catching will be advised for kids at the Easy Level.
 - Exercises for kids at the Medium Level will be a little more difficult, such as running and throwing a ball in various directions.
 - The system will suggest difficult activities that require balance, coordination, and motor control for kids at the Hard Level, such as kicking a ball, hand standing and dribbling
- To make sure the system adapts the task difficulty over time based on the child's progress, the child will receive feedback at every level. This maintains a suitable level of challenge to encourage skill improvement while guaranteeing ongoing development.
- By providing tailored suggestions, the system will increase the likelihood that kids will remain interested and advance their motor skills at a reasonable rate, which will improve their physical development.

- **Design a task recognition model to assess children's performance**

The system will use task recognition technology with OpenCV and MediaPipe to precisely evaluate a child's performance, which is essential for comprehending and analyzing the child's movements:

- Skeletal data will be extracted from the child's video input using MediaPipe Pose. The positions of the body's joints, including the elbows, knees, wrists, and ankles, are recorded by this skeleton data at every video frame.
- The system will monitor the child's movement patterns to determine whether they are following the right order. It will evaluate the child's ability to walk, jump, and balance, for instance.
- The system will compare the child's movements to a model trained on labeled data (i.e., video clips of correct actions) using OpenCV for video analysis. The degree to which the child's actions match the anticipated movements will determine the performance score.
- The system will give prompt feedback if the child's movements are inconsistent or incorrect, pointing out areas that need work and providing advice on how to do the movement more effectively.
- By tracking the child's performance in multiple sessions, the system can also track progress over time and provide parents, teachers, and therapists with visual progress reports and a thorough performance history.

2. Methodology

2.1. Methodology

2.1.1. Gross motor skills selection

It's critical to carefully choose exercises that are appropriate for children with Down syndrome's abilities and assess whether these activities are doable for them in order to help them develop their gross motor skills. We arranged multiple meetings with Dr. Sudarshi Lenagala, a pediatric specialist from Diana Princess of Wales Hospital in the United Kingdom, to make sure the chosen activities are suitable and successful. Expert knowledge regarding the kinds of gross motor skills that are most helpful for kids with Down syndrome, as well as how to evaluate their development and determine the right skill levels for each child, was made possible thanks in large part to these online sessions.

The children were divided into various levels in the project's initial phase according to their proficiency with fundamental gross motor tasks. In order to accomplish this, a basic gross motor skill was assigned to each child in order to evaluate their level of

proficiency. This strategy was created to make sure the kids' activities aren't too easy or too hard, which will keep them interested and inspire them to keep getting better.

In collaboration with Dr. Sudarshi, we determined that there are nine basic gross motor skills that fall into three progressive difficulty categories: Easy, Medium, and Hard. The physical difficulties and skills unique to children with Down syndrome—particularly their low muscle tone and other motor development issues—were taken into consideration when selecting these activities. The following is a breakdown of the activities by difficulty:

- **Easy Level:** This level covers fundamental skills like walking, standing, and catching. These easy exercises concentrate on basic movement patterns that kids with Down syndrome can begin practicing without much difficulty. The development of stability, balance, and coordination requires these motions.
- **Medium Level:** Activities at this level include throwing and running, which are a little more difficult. Strength, coordination, and motor control are the goals of these exercises. As kids run faster and perform more intricate movement patterns, running and throwing also aid in the development of spatial awareness and body coordination.
- **Hard Level:** The most difficult level involves more complex motor skills like kicking a ball, dribbling, and handstands. Increased muscle control, flexibility, and general physical strength are necessary for these activities. They combine dynamic movement patterns, balance, and accuracy to help children develop their motor skills and perform more complex, coordinated actions.

The viability and safety of each of these activities for kids with Down syndrome were thoroughly assessed. Their low muscle tone, which frequently makes it more difficult for them to carry out tasks requiring a great deal of strength or balance, was given particular attention. Every exercise's viability was evaluated to make sure it could be done without running the risk of overdoing it or getting hurt. We also carefully considered any modifications that might be required to the exercises in order to accommodate each child's unique needs and abilities. In addition to assisting in adjusting exercises to each child's skill level, this process of choosing suitable activities for each child guarantees that the activities are developmentally appropriate. The objective is to help kids develop their gross motor skills in a safe, interesting, and efficient manner by offering a steady increase in difficulty that encourages ongoing progress. We hope to establish an inclusive learning environment where all children with Down syndrome can engage in the activities and advance their motor development at their own pace by putting this structured approach into practice.

Dr Sudarshi Lenagala
Speciality doctor in Paediatrics
Diana Princess of Wales hospital
Grimsby
United Kingdom
Tel: 00947590733184
E mail: sudarshiwl@yahoo.com
02/04/2025

Research Panel
Supervisor and Co-supervisor
Department of software Engineering
SLIIT

To whom it may concern

The following actions and level of measurements are suitable for the children with Down syndrome aged 5-10 years old.

- 1/ 0-40 Easy level (Catch, stand, Walk)
- 2/ 41-70 Medium Level (Throw, Run)
- 3/ 80 >= Hard Level (Dribble, Handstand, Kick a ball)

Children with Down syndrome are delayed in development mile stones. A journal giving detailed information about the development was taken as the reference. The journal article is attached.

Thank you very much.
Yours sincerely



Dr Sudarshi Lenagala

Dr S Lenagala Gamaralalage
Speciality Doctor
GMC: 6106120
MBBS, MRCPCH
Diana Princess of Wales Hospital
Scartho Road, Grimsby DN33 2BA
sudarshi.lenagala@nhs.net
03033 304778

Figure 1: External Supervisor approval for Gross motor skills

Gross Motor Skills	Birth to 3 Years	3 – 5 years	5 – 8 Years
	Bottom shuffling	Delayed running	Inability to throw
	Fear climbing	Falling	Catch
			Not picked for team activities

Table 1: Some limitations of DS children

2.1.2. .Project Requirements

2.1.2.1. Functional Requirements

- Child should be able to get random video to identify the child level
- As ready for starting the process once the camera is open child should be in appear in the frame (easy to get all the key points to more accurate result)
- Compare the uploaded video with the trained model
- Provide the actual level of the child based on the child's performance
- Based on the level, provide the activities to improve their gross motor skills
- Provide accurate feedback for the users

2.1.2.2. Non-functional

- Interactive to the DS Children
- Easy to understand the process and use
- Accuracy
- Performance

2.1.2.3. Software Requirements

- Python
- ReatJS
- RNN
- LSTM
- Deep Learning
- Machine Learning
- Git
- CSS
- OpenCv
- MediaPipe
- Web Camera

2.1.3. Project Breakdown

2.1.3.1. Grant Chart

ID	Name	2024										2025					
		Mar 2024	Apr 2024	May 2024	Jun 2024	Jul 2024	Aug 2024	Sep 2024	Oct 2024	Nov 2024	Dec 2024	Jan 2025	Feb 2025	Mar 2025	Apr 2025	May 2025	Jun 2025
1	Initialization & Requirement Analysis																
2	Group Registration																
3	Topic Assessment Form																
4	Project Charter																
5	Proposal Presentation																
6	Proposal Reports																
7	Progress Presentation I																
8	Check List I																
9	Research Paper																
10	Progress Presentation II																
11	Final Reports																
12	Check List II																
13	Final Presentation																
14	Submission of RP																
15	Project Website																
16	Research Logbook																

Figure 2: Grant Chart

2.1.3.2. Work Breakdown Chart

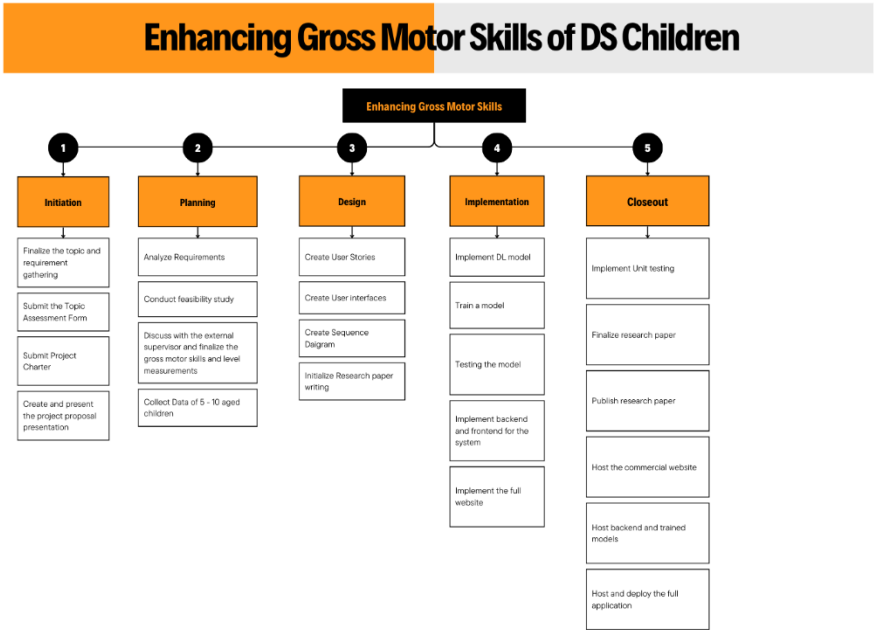


Figure 3: Work Breakdown Chart

2.1.5. Research Domain

This research mostly drives through following topics:

- Knowledge about Down Syndrome and motor skill development of DS children

Down syndrome, being a genetic condition caused by the presence of an extra chromosome 21, causes developmental delay, intellectual disability, and motor impairment.

Both the fine and gross motor skills in children with Down syndrome are affected, thus impacting their physical, cognitive, and social development [1].

Children with Down syndrome often experience delays in motor control due to differences in how their brain processes information. Their slower cognitive processing affects both fine and gross motor tasks. Your work, which focuses on using pose estimation and AI to analyze motor movements, is grounded in the understanding that these children may benefit from technology that supports their motor control and provides real-time feedback on their movements [3].

- DS children engagement in gross motor skill activities

Physical activity with Down syndrome children is crucial to develop motor skills and general health. Literature suggests that inducing physical activity can positively influence cognitive as well as motor development, creating stronger social participation and autonomy [2].

Physical exercise comes hand in hand with mental growth, particularly for Down syndrome children. Research has established that physical activity through motor skill like walking, catching, or jumping has the ability to play a constructive role in cognition like attention span and memory [6]

- Existing technological approaches

Advances in machine learning and computer vision technology have brought new possibilities for assessing and improving motor capacity in children with disabilities. Detecting human movement by tracking movement via key points (skeleton-based data) [4]

Artificial intelligence and machine learning techniques, including deep learning models, are revolutionizing motor skills assessment. Deep learning models such as Graph Convolutional Networks (GCNs) have been employed in current studies to assess the gross motor functioning of toddlers with impairments [11].

- Existing approaches with video processing

Research has shown that video modeling, where children watch videos of motor activities being performed, is an effective method for teaching Down syndrome children. Determines the effectiveness of teaching basketball movements through video modeling in Down syndrome individuals. [10]

2.2. Commercialization

This research project focused on stakeholders who are children with Down Syndrome who want to develop gross motor skills with the support of their parents

Target Audience

The target audience of this approach is those who are working directly with children with Down Syndrome

Children with Down Syndrome

Special Education teachers

DS children's parents and caregivers

Child Health Care Systems

2.3. Testing and Implementation

2.3.1. Collecting video inputs for dataset

Used Python Libraries and advantages

- MediaPipe: Pose Estimation with real-time and pretrained models
- OpenCV: Provide a robust set of computer vision tools
- NumPy: Efficient numerical computations
- Os: simplifies the file system

```
import cv2
import mediapipe as mp
import numpy as np
import os
import time

mp_pose = mp.solutions.pose
mp_drawing = mp.solutions.drawing_utils
pose = mp_pose.Pose(static_image_mode=False, min_detection_confidence=0.5, min_tracking_confidence=0.5)
```

Figure 5: Libraries in used in Collect.py

MediaPipe Pose Estimation:

The Google-developed MediaPipe Pose software package provides a method of analyzing the overall body pose and movement while broadcasting or exercising. Here, by finding pose landmarks, various body joints are aggregated together to create a whole pose. The software, therefore, is used to find key points (such as face, neck, faces of friends).

```

data_dir = "collected_data"
if not os.path.exists(data_dir):
    os.makedirs(data_dir)

actions = ["catch", "walk", "stand"]
for action in actions:
    if not os.path.exists(os.path.join(data_dir, action)):
        os.makedirs(os.path.join(data_dir, action))

cap = cv2.VideoCapture(0)

# Set camera to 1280x720 (common 16:9 resolution)
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)

def extract_keypoints(results):
    if results.pose_landmarks:
        return np.array([[lm.x, lm.y, lm.z] for lm in results.pose_landmarks.landmark]).flatten()
    else:
        return np.zeros(33 * 3)

action_index = 0
sequence_index = 0
frame_index = 0
keypoints_data = []
collecting = False

landmark_drawing_spec = mp_drawing.DrawingSpec(color=(0, 255, 0), thickness=3, circle_radius=5)
connection_drawing_spec = mp_drawing.DrawingSpec(color=(255, 0, 0), thickness=4)

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

```

Figure 6: MediaPipe

In the above code, WebCam or video stream of children exercising target scene, pose landmark detection, and dynamic motion searching (e.g., walking, catching, standing). Lastly, the collected key points (data) are saved in.npy files.

OpenCV:

The OpenCV (Open-Source Computer Vision Library) software library includes a variety of visual processing capabilities. This code, written with OpenCV, allows video capture and other operations when using a WebCam or video stream.

Automatic data collection:

In the code, detailed key points data are collected at 30 frames per second (FPS) when running time limits and chapters. In this case, the code records the actions performed (e.g., "catch", "walk", "stand").

```

if collecting:
    keypoints = extract_keypoints(results)
    keypoints_data.append(keypoints)

    if frame_index % 30 == 0 and frame_index != 0:
        save_path = os.path.join(data_dir, actions[action_index], f"sequence_{sequence_index}.npy")
        np.save(save_path, np.array(keypoints_data))
        keypoints_data = []
        sequence_index += 1

```

Figure 7: Extract Key points

When collecting data, suggesting or having suggested actions are provided, using an action classification approach.

```
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.flip(frame, 1)

    height, width = frame.shape[:2]

    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    results = pose.process(frame_rgb)

    black_background = np.zeros(frame.shape, dtype=np.uint8)
    if results.pose_landmarks:
        mp_drawing.draw_landmarks(
            frame,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS,
            landmark_drawing_spec=landmark_drawing_spec,
            connection_drawing_spec=connection_drawing_spec
        )
        mp_drawing.draw_landmarks(
            black_background,
            results.pose_landmarks,
            mp_pose.POSE_CONNECTIONS,
            landmark_drawing_spec=landmark_drawing_spec,
            connection_drawing_spec=connection_drawing_spec
        )
```

Figure 8: Convert BRG into RGB

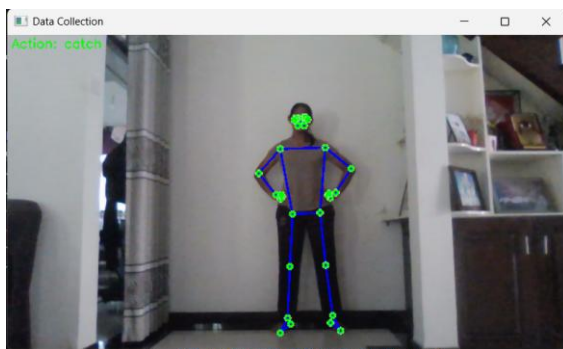


Figure 10: Man with Skeleton

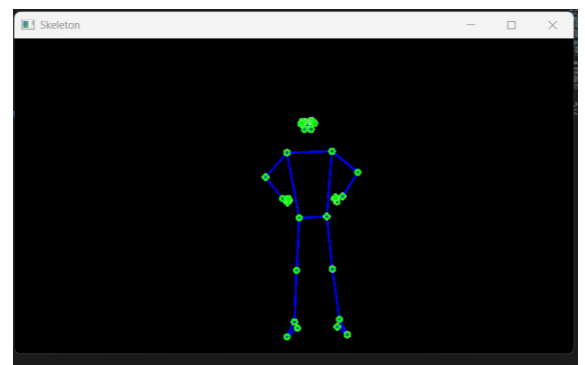


Figure 9: Same man with Black background

Figure

2.3.2. Data Preprocessing

The key function of this Python code is to convert the human motion data that has been collected into a desired set of skeleton sequences. This helps in giving a

sufficient, clean, and suitable dataset for machine learning algorithms. The processing process involves some basic steps, which involve frame selection corresponding to the structure, length correction, and formation of the dataset. These are necessary to recognize moods, assess performance, and study motor skills as a whole.

Choosing Proper Skeleton Frames:

A function called `is_valid_skeleton()` is used here. It performs a validity test by sorting the x and y coordinate values of each frame according to the number of non-zero values. The non-zero values of the x and y coordinates of the 33 keypoints are calculated, and if the number is equal to or greater than a threshold (default 10), the frame is valid. This process serves to discard the invalid frames and retain only the frames that can be used further.

```
def is_valid_skeleton(sequence, threshold=10):
    valid_frames = []
    for frame in sequence:
        non_zero_count = np.count_nonzero(frame[:, :2])
        if non_zero_count >= (33 - threshold) * 2:
            valid_frames.append(frame)
    return np.array(valid_frames) if len(valid_frames) > 0 else None
```

Figure 11: Find the skeleton

Padding Sequence Length: The second step is to apply the `pad_sequences()` function to standardize the length of each sequence. This is necessary in order to transform the input data into an adequate length to present to a machine learning algorithm. Zero padding is added to a sequence if it is shorter than the maximum length, and the longer sequences are truncated. This prevents very long sequences from being shown to the model as data.

```
def pad_sequences(keypoints_data, max_length):
    padded_data = []
    for sequence in keypoints_data:
        if len(sequence) < max_length:
            padded_sequence = np.pad(sequence, ((0, max_length - len(sequence)), (0, 0)), mode='constant')
        else:
            padded_sequence = sequence[:max_length]
        padded_data.append(padded_sequence)
    return np.array(padded_data)
```

Figure 12: Reshape the dataset

Walk through the dataset and reshape the data:

Inside the `preprocess_dataset()` function, the .npy copies for each action class (easy, medium, hard) are processed by iterating over the dataset directory structure. Inside, each .npy file is read and reshaped, and the `is_valid_skeleton()` function is called to select valid frames. Serial data is reshaped, and the vectors for the key points are added to that shape.

Action labels (easy, medium, hard) and processed key points are concatenated and saved in another list. Together, the current sequence length is computed, and padding or truncation is performed to the remaining sequences.

```
def preprocess_dataset(dataset_dir, output_file):
    labels = []
    keypoints_data = []
    max_length = 0

    for action in os.listdir(dataset_dir):
        action_dir = os.path.join(dataset_dir, action)
        if not os.path.isdir(action_dir):
            continue

        for sequence_file in os.listdir(action_dir):
            if sequence_file.endswith(".npy"):
                sequence_path = os.path.join(action_dir, sequence_file)
                keypoints = np.load(sequence_path)
                keypoints = keypoints.reshape(-1, 33, 3) if keypoints.shape[1] == 99 else keypoints
                filtered_keypoints = is_valid_skeleton(keypoints)
                if filtered_keypoints is not None:
                    filtered_keypoints = filtered_keypoints.reshape(filtered_keypoints.shape[0], -1)
                    keypoints_data.append(filtered_keypoints)
                    labels.append(action)
                    max_length = max(max_length, len(filtered_keypoints))

    keypoints_data = pad_sequences(keypoints_data, max_length)
    np.savez(output_file, keypoints=keypoints_data, labels=np.array(labels))

preprocess_dataset("collected_data", "preprocessed_easy_data.npz")
```

Figure 13: Save file with .npz

Saving the Preprocessed Data:

After preprocessed all the sequences, the dataset created by key points and labels was saved into a .npz file using the np.savez() function. The .npz file format is very useful in storing and loading big datasets. This preprocessed dataset can be utilized as input data for machine learning models.

```
max_length = max(max_length, len(filtered_keypoints))

keypoints_data = pad_sequences(keypoints_data, max_length)
np.savez(output_file, keypoints=keypoints_data, labels=np.array(labels))

preprocess_dataset("collected_data", "preprocessed_easy_data.npz")
```

Figure 14: Save the dataset

Demonstration:

Finally, this code is a procedure of transforming raw skeletal motion data to a comprehensive and reliable dataset. By eliminating invalid frames, padding sequences, reshaping the data, and saving the dataset, raw motion data is transformed into a clean, organized, and optimized dataset. All these procedures constitute a dataset that can be used in machine learning models, action recognition, motor skill assessment, and other purposes.

v collected_data	
	v catch
	sequence_0.npy
	sequence_1.npy
	sequence_2.npy
	sequence_3.npy
	sequence_4.npy
	sequence_5.npy
	sequence_6.npy
	sequence_7.npy
	sequence_8.npy
	sequence_9.npy
	sequence_10.npy
	sequence_11.npy
	sequence_12.npy
	sequence_13.npy
	sequence_14.npy
	sequence_15.npy
	sequence_16.npy
	sequence_17.npy
	sequence_18.npy
	sequence_19.npy
	sequence_20.npy
	sequence_21.npy
	sequence_22.npy
	sequence_23.npy
	sequence_24.npy
	sequence_25.npy
	sequence_26.npy
	sequence_27.npy
	sequence_28.npy

Figure 15: Collected .npy dataset

2.3.3. Train Model

From a pre-processed human body key points dataset, create a deep learning model that classifies from key points. To achieve this, we first import the required libraries in code. For example, numpy for simplifying operations in large data arrays, torch (PyTorch) for tensor computation, model definition, and training. Sklearn to divide the dataset into train-test and calculate evaluation metrics. Matplotlib.pyplot to display data visualizations (graphs/charts).

```
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from sklearn.model_selection import train_test_split
from torch.utils.data import DataLoader, TensorDataset
from sklearn.metrics import classification_report, confusion_matrix
import matplotlib.pyplot as plt
from sklearn.utils.class_weight import compute_class_weight
```

Figure 16: Libraries used in training

The preprocessed_easy_data.npz compressed numpy file is imported next. The key points (child or trained subject's body-bound coordinates) and labels (easy, medium, hard) within the file are imported and assigned to the last two variables. The key points are the model input data, and the labels are the model output targets.

Then, the text values of the labels (easy, medium, hard) are converted to integer values and are converted to an acceptable form to be input to the machine learning model. To it, a dictionary (label_to_int) is created and converted to. In reality, the neural network requires numerical data type inputs (e.g., 0, 1, 2).

Then print the class balance of the data and there is a risk of imbalance in those labels. For example, if there are more hard labels than easy labels, then there could be a risk of bias in the model. So, this analysis checks for the balance, and if it is not right, then one can apply the techniques of data augmentation to correct the imbalance.

```
data = np.load("preprocessed_easy_data.npz")
keypoints = data["keypoints"]
labels = data["labels"]

label_to_int = {label: i for i, label in enumerate(np.unique(labels))}
int_to_label = {v: k for k, v in label_to_int.items()}
labels = np.array([label_to_int[label] for label in labels])

|
unique, counts = np.unique(labels, return_counts=True)
print("Class distribution:", dict(zip([int_to_label[u] for u in unique], counts)))
```

Figure 17: Label the classes

Then, the key points and labels are separated into a train set and a test set. Since this is a stratified split, it is ensured that each label class has an equal presence on either side (train/test). In this data split, 80% is used for the training set and 20% for the testing set.

```
x_train, x_test, y_train, y_test = train_test_split(keypoints, labels, test_size=0.2, random_state=42, stratify=labels)
```

Figure 18: Train set and Test set

Then, the training data and testing data retrieved are converted into tensors (PyTorch tensor objects) that are easy to input into the model. TensorDataset and DataLoader are used for efficient model training. DataLoader with a batch size of and with random shuffling (shuffle=True) enabled for training data is used, which improves training diversity and model generalization capability.

Classification neural network model is constructed with an input layer, several hidden layers, and a final output layer. A loss function is established, and an optimizer (Adam Optimizer) is used to optimize the learning.

```
x_train = torch.tensor(x_train, dtype=torch.float32)
x_test = torch.tensor(x_test, dtype=torch.float32)
y_train = torch.tensor(y_train, dtype=torch.long)
y_test = torch.tensor(y_test, dtype=torch.long)

train_dataset = TensorDataset(x_train, y_train)
test_dataset = TensorDataset(x_test, y_test)
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)

class ActionRecognitionModel(nn.Module):
    def __init__(self, input_size, hidden_size, num_classes):
        super(ActionRecognitionModel, self).__init__()
        self.lstm1 = nn.LSTM(input_size, hidden_size, batch_first=True, bidirectional=True)
        self.lstm2 = nn.LSTM(hidden_size * 2, hidden_size // 2, batch_first=True, bidirectional=True)
        self.fc = nn.Linear(hidden_size, num_classes)

    def forward(self, x):
        h_lstm1, _ = self.lstm1(x)
        h_lstm2, _ = self.lstm2(h_lstm1)
        h_lstm2 = h_lstm2[:, -1, :]
        out = self.fc(h_lstm2)
        return out
```

Figure 19: Training Dataset

Model is trained across 50 training epochs. In each training

epoch, performance measures such as training loss values and training accuracy are monitored, and the progress of the model is monitored continually.

```
for epoch in range(50):
    model.train()
    epoch_loss = 0
    for inputs, targets in train_loader:
        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, targets)
        loss.backward()
        optimizer.step()
        epoch_loss += loss.item()

    train_losses.append(epoch_loss / len(train_loader))

    model.eval()
    correct = 0
    total = 0
    all_targets = []
    all_predictions = []
    with torch.no_grad():
        for inputs, targets in test_loader:
            outputs = model(inputs)
            _, predicted = torch.max(outputs.data, 1)
            total += targets.size(0)
            correct += (predicted == targets).sum().item()
            all_targets.extend(targets.cpu().numpy())
            all_predictions.extend(predicted.cpu().numpy())

    accuracy = 100 * correct / total
    val_accuracies.append(accuracy)
    scheduler.step(epoch_loss)
```

Figure 20: Use 50 epochs

Once trained, the model is able to be tested on the test set. In this instance, accuracy, precision, recall, and F1-score evaluation metrics are computed to examine the classification performance of the model. For the rest, a confusion matrix is built, a prediction error visualization is performed, and trends in which the model incorrectly predicts most of its errors are also shown as support. This model is highly trained, optimized machine learning model that can efficiently label easy, medium, and hard classifications to future key point inputs such as live movements as videos

Different logics I used for training and there advantages

- Stratified Shuffle Split: split the dataset, we do maintain the original class proportion. That is, the proportion of classes like easy, medium, and hard is the same in the train set as well as in the test set. These supports having an unbiased test or training when we train or test the model.
- PyTorch tensorDataset and DataLoader: In general, if each data point is fed into the model sequentially, the training process becomes extremely slow. Additionally, wasteful memory usage is also encountered.

Batch processing is thus imagined. Batch processing is the process of splitting the data into small batches of files and adding each batch to the model at once. This is performed so that the model can be processed in parallel. The batch size is chosen as a particular value, such as 32, for the purpose of maximizing the training efficiency.

TensorDataset:

Unique PyTorch object that constructs a single dataset object from input labels and features. It's passed to the DataLoader.

The DataLoader: batches, shuffles, and efficiently loads data

loading automated in PyTorch. Each batch is created and fed into the model if a DataLoader is utilized, which leads to less memory waste and less training time.

Loss Function: The Loss Function is very vital while training a machine learning model. The Loss Function computes the difference between the model's prediction (guess) and the actual answer (true label). That is, the loss function indicates how well or poorly the model predicts.

If the Loss value is big, the model does not make accurate predictions, there is a large difference between the predictions and the correct answers.

Loss value is small: the model can be said to predict accurately

```
crit = nn.CrossEntropyLoss(weight=class_weights)
optimizer = optim.Adam(model.parameters(), lr=0.001)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer, 'min', patience=5, factor=0.1)
```

Figure 21: Loss function, Data Loader

Loss function that is suitable for a multi-class classification problem. We have a classification problem where we want to predict labels like easy, medium, and hard, so Cross Entropy Loss is a perfect fit.

Optimizer: An optimizer is an algorithm that helps reduce the loss function by adjusting the model's weights to their optimal values during training a machine learning model

When the model is trained, it attempts to make better predictions so that the loss can be minimized. The optimizer utilizes gradients to do so. Gradients are feedback on how the loss is changing with respect to the weights. Utilizing this feedback, the model attempts to minimize the loss value by adjusting or changing its weights.

Adam Optimizer: An intelligent optimizer that has rapid convergence and

adaptive learning features, mostly utilized in recent deep learning frameworks.

- Epochs: Epochs are the time instances when the entire training set is inputted into the model. To put it simply, an epoch is the time when the model has viewed all data points.

1 epoch is $\rightarrow$ the time the model has learned from each example in the dataset.

100 epochs is $\rightarrow$ the time the model has learned 100 times from the dataset.

That is because the model is trained to reduce gradually the loss and increase the prediction accuracy. The model is slightly improved every epoch.

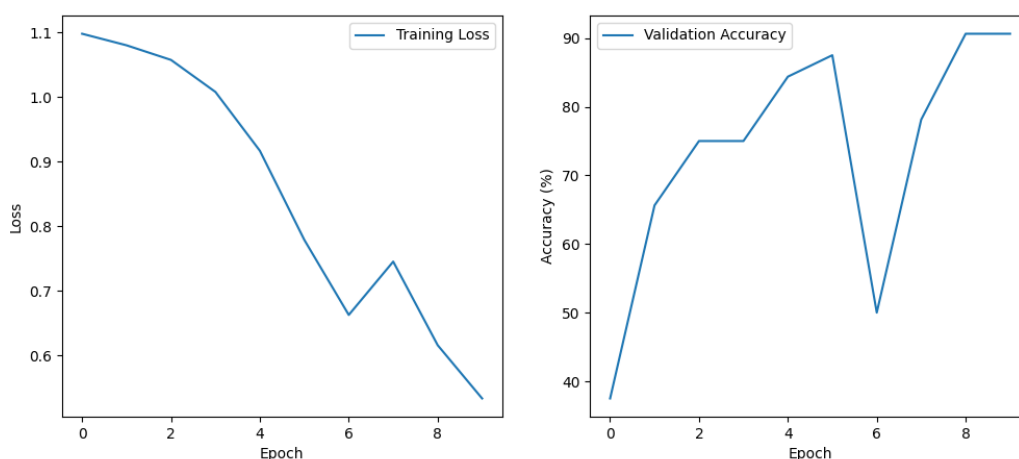


Figure 22: Model accuracy and loss

2.3.4. Developing gross motor skills research component process

A dynamic learning process that is intended to assist the child in developing fundamental motor skills through an interesting, interactive system will be presented to the parent or child when they navigate to the section titled "Develop Gross Motor Skills." The child will be given a randomly generated gross motor skill task, like standing, walking, or jumping, as soon as they enter this section. These activities are intended to serve as a springboard for the child's physical development and are selected according to their applicability to the child's present developmental stage.

The child will be asked to carry out the specified action, such as standing, walking, or jumping, in a manner that represents their present abilities as part of this preliminary procedure. In addition to encouraging the child to participate in the

activity, this pragmatic approach enables the system to evaluate the child's innate performance. The child will use a basic video recording feature in the system to record their action after attempting the task. After that, the recorded video will be added to the system for additional examination.

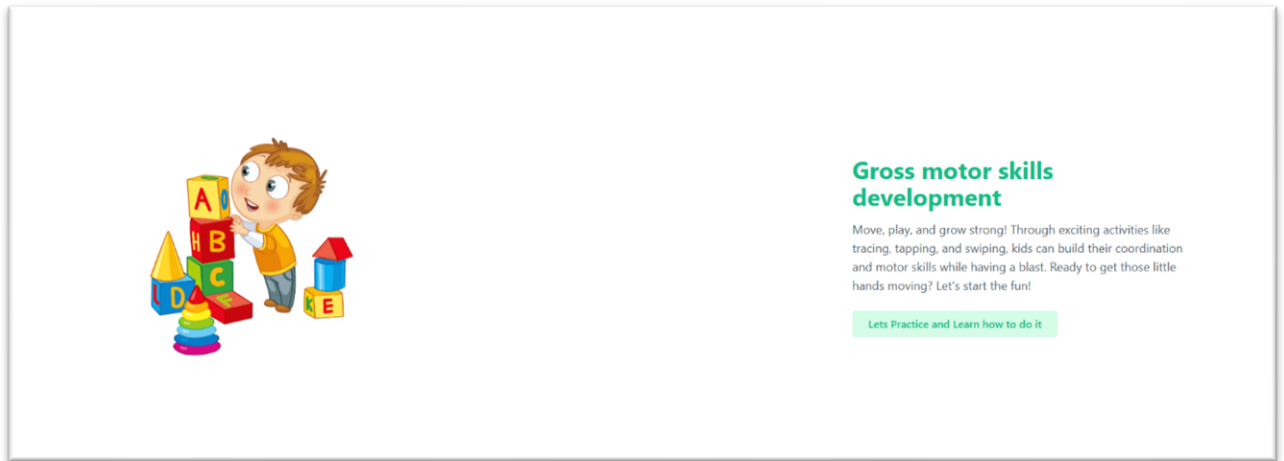


Figure 23: Frontend Interface

Figure

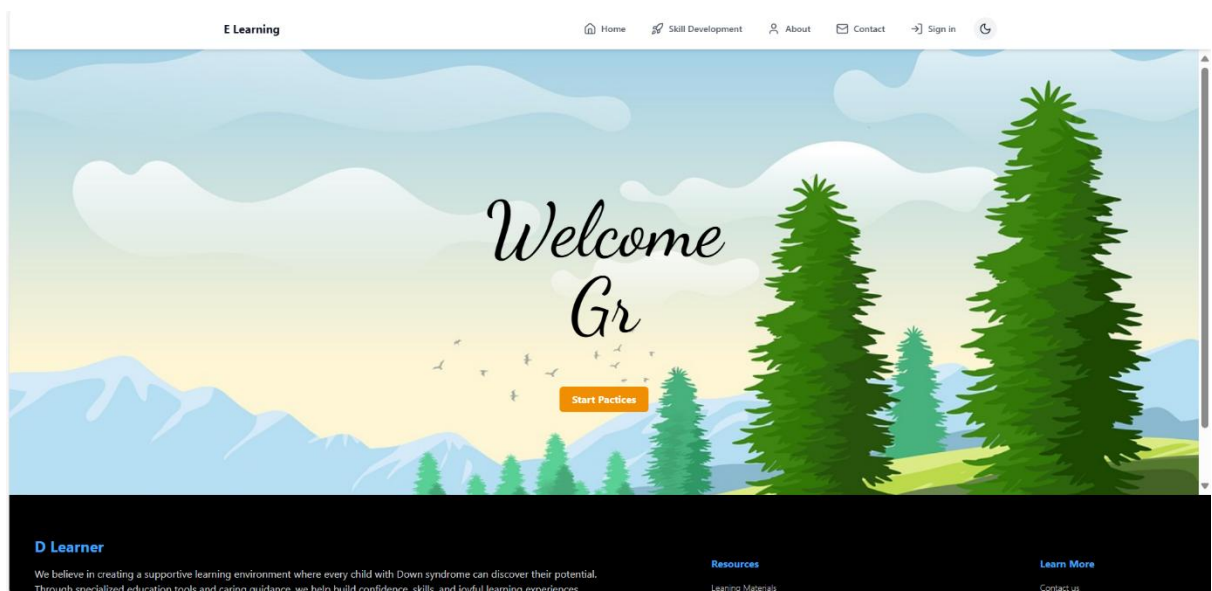


Figure 24: When click on the gross motor skill

This screen will start with a sound, and the child can start to learn and practice gross motor skills in here

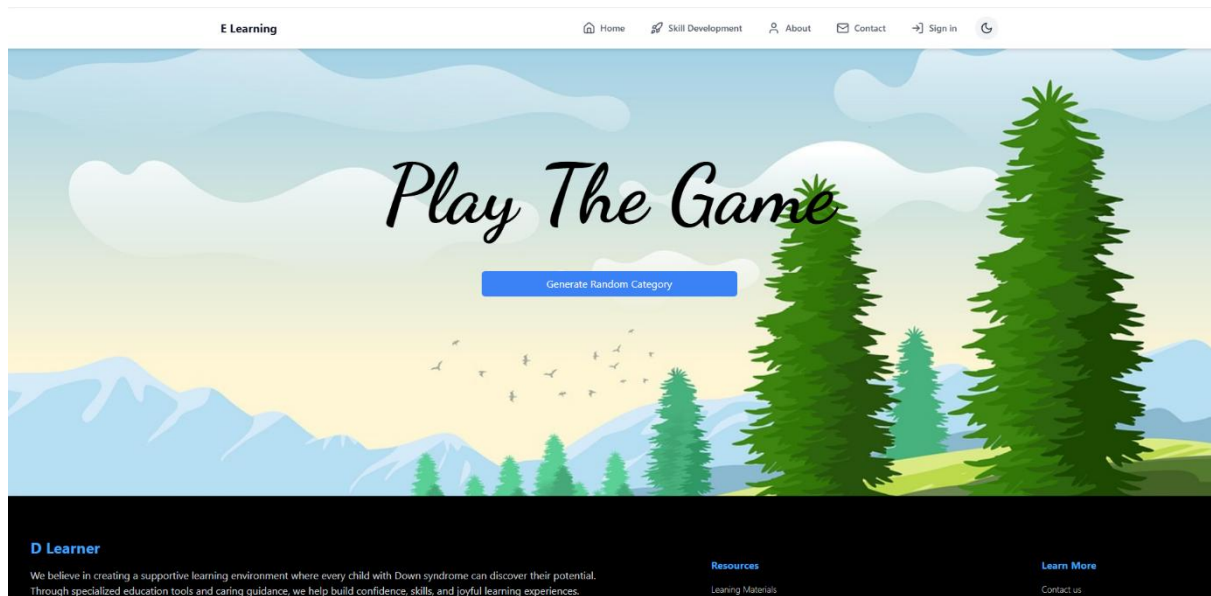


Figure 25: Generate random video

By using that screen, a child can generate a random video to perform their initial stage and identify his/her level.

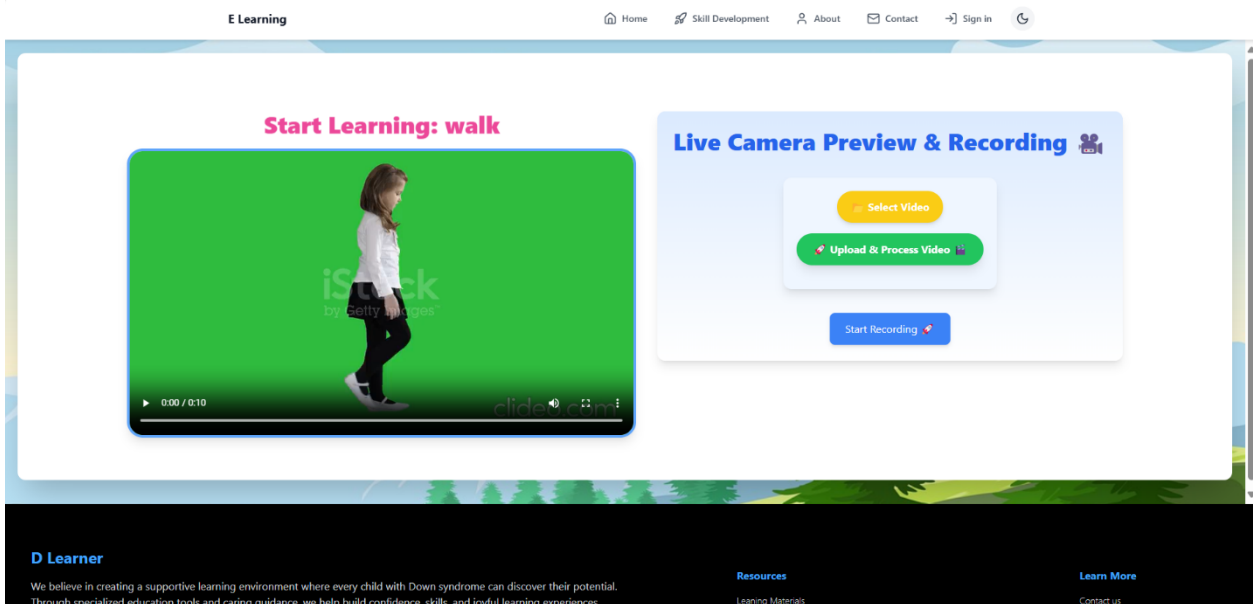


Figure 26: Generate Random video

Figure

2.3.5. Categorize into levels

A three-layered system architecture has been developed to achieve robust action recognition. The architecture begins with two stacked Long Short-Term Memory (LSTM) layers which perform essential functions for decoding body movement sequences. LSTM layers feature a hidden size value of 64 which ensures an equilibrium between model complexity and computational efficiency.

```
easy_classes = 3
medium_classes = 2
hard_classes = 3

class ActionRecognitionModel(torch.nn.Module):
    def __init__(self, input_size, hidden_size, num_classes):
        super(ActionRecognitionModel, self).__init__()
        self.lstm1 = torch.nn.LSTM(input_size, hidden_size, batch_first=True, bidirectional=True)
        self.lstm2 = torch.nn.LSTM(hidden_size * 2, hidden_size // 2, batch_first=True, bidirectional=True)
        self.fc = torch.nn.Linear(hidden_size, num_classes)
```

Figure 27: Level categorization

The application of two LSTM layers demonstrates high effectiveness for analyzing complex activities which include throwing and dribbling as well as dynamic poses that require intricate coordination. This two-layer system enables bidirectional learning by processing information in both directions which improves the system's ability to capture temporal dependencies and boosts prediction accuracy.

The initial LSTM layer functions to discover small-scale detailed patterns throughout the movement sequences. It extracts detailed motion features from specific regions, which serve as fundamental components for decoding complicated actions. The second LSTM layer merges these fine-grained patterns into an advanced interpretation which captures the full meaning and sequence of the performed action. The hierarchical learning structure equips the system to generate predictions that are both better informed and more dependable.

The processed output from the LSTM layers undergoes transformation into a basic fully connected linear layer. At the end of the processing pipeline this linear layer functions as the classifier, which determines the specific action prediction.

The system takes key points extracted from child movements as its input. The system utilizes the selected model to process key points which have been classified into difficulty levels of easy, medium or hard following an initial assessment. The model processes the key points before generating raw output scores. The system transforms these raw output scores into probability values to make decisions about the child's performance level in a probabilistic manner.

```
def extract_keypoints(video_path):
    pose = mp_pose.Pose(static_image_mode=False, min_detection_confidence=0.5, min_tracking_confidence=0.5)
    cap = cv2.VideoCapture(video_path)
    keypoints_list = []
    frames_with_keypoints = 0
    total_frames = 0

    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break
        total_frames += 1
        frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        results = pose.process(frame_rgb)
        if results.pose_landmarks:
            frames_with_keypoints += 1
            keypoints = np.array([[lm.x, lm.y, lm.z] for lm in results.pose_landmarks.landmark]).flatten()
            keypoints_list.append(keypoints)

    cap.release()
    return np.array(keypoints_list), frames_with_keypoints, total_frames
```

Figure 28: Extract Key points

Using the calculated probabilities:

If the child's movement displays clear difficulty or also differs significantly from the expected form and the probability value lies between 1% to 40%, the child will be categorized into the Easy level.

In case he performs moderately, it means he did capture some aspect of the movement correctly, but the performance is not altogether precise; the probability lies between 41% to 79%, and the child is assigned to the medium level.

If the child does the movement just as we are expecting (with high certainty), with a probability score of 80% or above. The child will be promoted to Hard level.

With this flexible assessment structure, each child is directed to the stage that suits his/her aptitude, thus facilitating and enhancing the engagement, skill growth of every child and boosting motivation to learn.

```
def get_message(score, detection_ratio):
    if detection_ratio < 0.1:
        return "No human detected in most frames. Please ensure you're in front of the camera."
    if score == 0:
        return "Action not recognized. Please try again."
    elif 1 <= score <= 10:
        return "It is very hard to find the action."
    elif 11 <= score <= 40:
        return "Child is finding it hard to perform the action. Please repeat the action as shown in the video and try again."
    elif 41 <= score <= 79:
        return "Child is performing at a medium level. Keep practicing to improve."
    else:
        return "Child is performing well! The action is easy for them."

def get_diff_level(score, detection_ratio):
    if detection_ratio < 0.1:
        return "No detection"
    if score == 0:
        return "No level"
    elif 1 <= score <= 40:
        return "Easy"
    elif 41 <= score <= 79:
        return "Medium"
    else:
        return "Hard"
```

Figure 29: Level Selection Percentage

```
if detection_ratio < 0.1:
    return jsonify({
        "predictions": [],
        "percentage": "0",
        "message": get_message(0, detection_ratio),
        "level": get_diff_level(0, detection_ratio),
        "predicted_action": "none",
        "actual_class": actual_class,
        "receivedPredictions": {},
        "detection_ratio": f"{detection_ratio:.2f}"
    })
```

Figure 30: Output

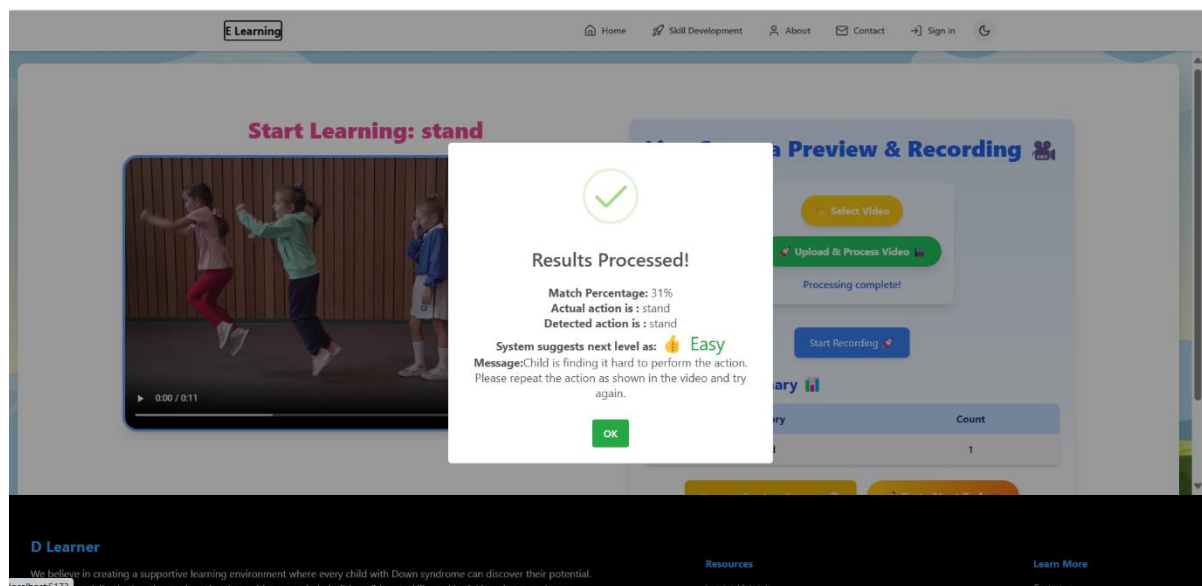


Figure 31: Child select to Easy Level

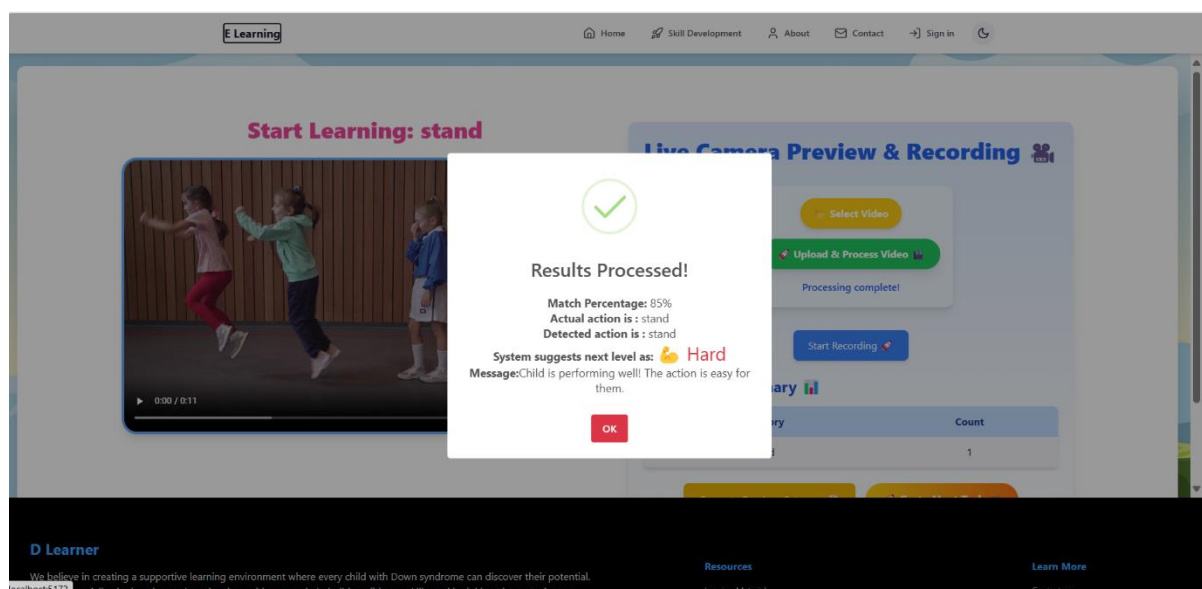


Figure 32: Child Select to Hard Level

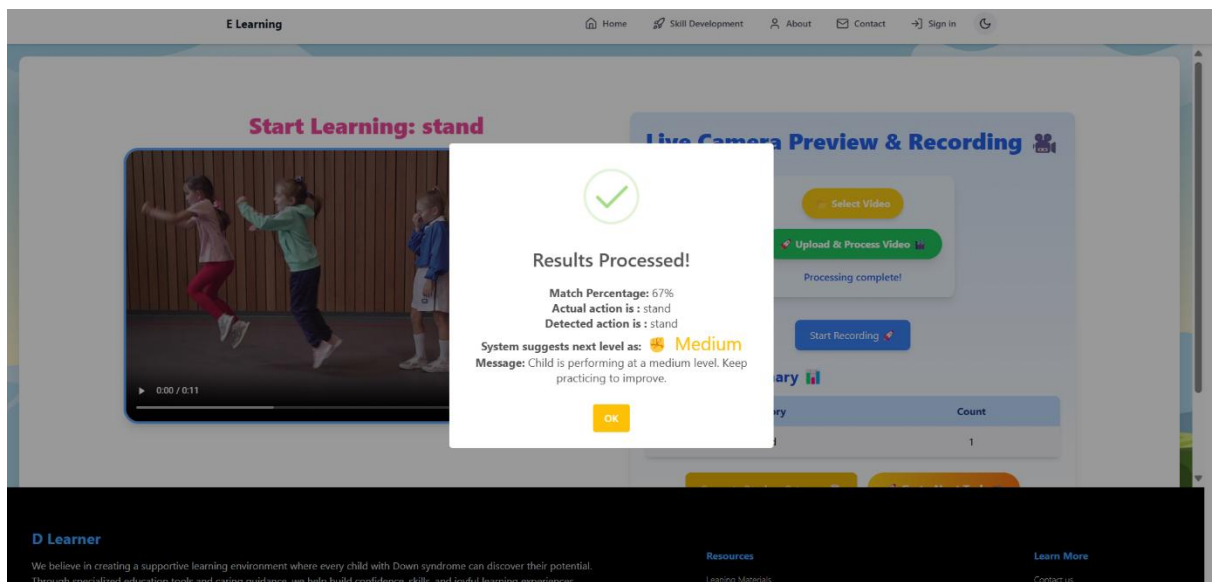


Figure 33: Child select to Medium Level

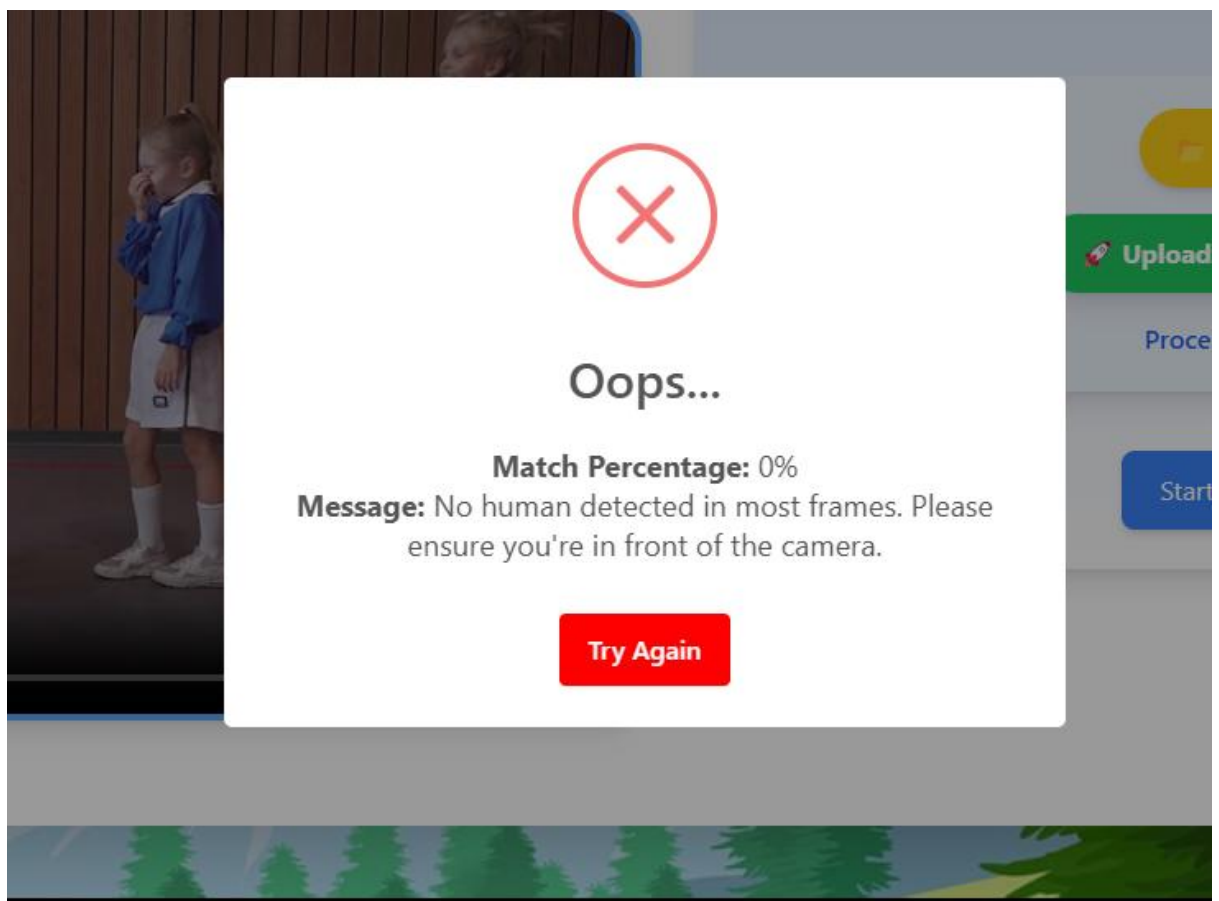


Figure 34: No human detected

2.3.6. Provide gross motor skills based on the selected level

After the analysis of bodily movements and probabilities, the system assigns easy, medium or hard skill levels to the child, and once the system evaluates the initial skill level of a child, the child will have a personalized set of gross motor skills tasks according to the assigned skill level. These tasks are tailored to the child's competencies, providing enough challenge to support growth without risk of frustration or loss of interest.

For every assigned gross motor skill, an instructional video acts as a visual guide. Using an application interface, the kid can generate and access this video. Well, these videos show you precisely how the movement intended to target is intended to be done and show you the movement that you should perform in incremental steps. The example videos provided demonstrate what to do, which includes the position, coordination, and timing of the action. After viewing, the child is directed to use what they have seen and try the task himself/herself. Children also learn through a process of observation and imitation, so these small interactions are key to developing and fine-tuning gross motor skills, particularly for those who thrive on strong visual prompts and demonstration.

The system not only provides the instructional support needed but also stimulates independent learning, and the child can learn at their own pace. With repeated practice and feedback over time, the child can improve their performance, gradually moving on to more complicated tasks as their ability increases.

2.3.7. Test Cases

2.3.7.1. GMS01

Field	Information
Test ID	GMS01
Test Scenario	Sign in with valid credentials
Test description	Verify that user can successfully log in with valid credentials
Test Data	Valid username and password
Test Steps	Navigate to login page > Enter valid username > Enter valid password > Click “Sign In” button
Expected Result	User successfully authenticated and directed to the home dashboard
Actual result	User successfully authenticated and directed to home dashboard as expected
Status	Pass
Priority	High

Table 2: Test Case – GMS01

2.3.7.2. GMS02

Field	Information
Test ID	GMS02
Test Scenario	Sign in with invalid credentials
Test Description	Verify that the system rejects login attempts with invalid credentials
Test Data	Invalid username and password
Test Steps	Navigate to login page > Enter valid username > Enter valid password > Click “Sign In” button
Expected Result	Authentication fails error message: "Invalid username or password"
Actual result	Authentication failed error message: "Invalid username or password"
Status	Pass
Priority	High

Table 3: Test Case - GMS02

2.3.7.3. GMS03

Field	Information
Test ID	GMS03
Test Scenario	Attempt to upload without selecting video
Test Description	Verify that the system prevents uploads when no video is selected
Test Data	No video selected
Test Steps	Navigate to Video Upload Section > Click “Upload: button without selecting a video
Expected Result	System displays notification: "Please select or record a video first!"
Actual result	System displayed notification: "Please select or record a video first!"
Status	Pass
Priority	Medium

Table 4: Test Case – GMS03

2.3.7.4. GMS04

Field	Information
Test ID	GMS04
Test Scenario	Upload recorded video
Test Description	Verify that recorded videos can be successfully uploaded
Test Data	Recorded video using system camera
Test Steps	Record video > Click “Upload” button
Expected Result	System displays "Uploading and processing..." notification > Video is successfully uploaded and processed

Actual result	The system displayed "Uploading and processing..." but upload failed after 30 seconds
Status	Fail
Priority	High

Table 5: Test Case – GMS04

2.3.7.5. GMS05

Field	Information
Test ID	GMS05
Test Scenario	Upload video with valid format
Test Description	Verify that videos with valid format can be uploaded successfully
Test Data	Valid .mp4 video file of child performing action
Test Steps	Navigate to Upload > Select valid .mp4 file > Click "Upload"
Expected Result	Video uploads successfully and is sent for processing
Actual result	Video uploaded successfully and processing started
Status	Pass
Priority	High

Table 6: Test Case – GMS05

2.3.7.6. GMS06

Field	Information
Test ID	GMS06
Test Scenario	Upload video with invalid format
Test Description	Verify that system rejects invalid file formats
Test Data	Invalid file formats (.png, .pdf)
Test Steps	Navigate to Upload > Select invalid .png, .pdf file > Click "Upload"
Expected Result	Video uploads successfully and is sent for processing
Actual result	Video uploaded successfully and processing started
Status	Pass
Priority	High

Table 7: Test Case – GMS06

2.3.7.7. GMS07

Field	Information
Test ID	GMS07
Test Scenario	Full body action detection
Test Description	Verify system correctly detects skeleton key points with full body visible
Test Data	Child with full body visible
Test Steps	Navigate to "Start Recording" > Position child with full body visible > Perform action > Click "Upload"
Expected Result	System successfully detects all skeleton key points and provides accurate feedback

Actual result	System detected all skeleton key points and provided accurate feedback
Status	Pass
Priority	High

Table 8: Test Case – GMS07

2.3.7.8. GMS08

Field	Information
Test ID	GMS08
Test Scenario	Partial body action detection
Test Description	Verify system behavior when only part of the body is visible
Test Data	Child with partial body visible
Test Steps	Navigate to “Start Recording” > Position child with partial body visible > Perform action > Click “Upload”
Expected Result	System detects available key points and provides feedback based on partial detection
Actual result	System detected some key points but feedback was inconsistent
Status	Partial Pass
Priority	Medium

Table 9: Test Case – GMS08

2.3.7.9. GMS09

Field	Information
Test ID	GMS09
Test Scenario	Extremely close camera positioning
Test Description	Verify system behavior when subject is positioned too close to camera
Test Data	Child positioned very close to camera
Test Steps	Navigate to “Start Recording” > Position child very close to the camera > Perform action > Click “Upload”
Expected Result	System displays notification: "Action not recognized. Please try again." due to insufficient key points detected
Actual result	System displayed notification: "Action not recognized. Please try again."
Status	Pass
Priority	Medium

Table 10: Test Case – GMS09

2.3.7.10. GMS10

Field	Information
-------	-------------

Test ID	GMS10
Test Scenario	Detection with environmental objects
Test Description	Verify system can filter out non-human objects in frame
Test Data	Child with multiple non-human objects in frame
Test Steps	Navigate to “Start Recording” > Position child with multiple objects > Perform action > Click “Upload”
Expected Result	System filters out non-human objects and correctly focuses on human detection only
Actual result	System correctly ignored non-human objects and focused on human detection
Status	Pass
Priority	Medium

Table 11: Test Case – GMS10

2.3.7.11. GMS11

Field	Information
Test ID	GMS11
Test Scenario	No human in video frame
Test Description	Verify system behavior when no humans are present in video
Test Data	Video with no humans present
Test Steps	Navigate to “Start Recording” > Record with no human > Perform action > Click “Upload”
Expected Result	System displays notification: "No human detected in most frames. Please ensure you're in front of the camera."
Actual result	System displayed: "No human detected in most frames. Please ensure you're in front of the camera."
Status	Pass
Priority	Medium

Table 12: Test Case – GMS11

2.3.7.12. GMS12

Field	Information
Test ID	GMS12
Test Scenario	Correct action recognition - Walk
Test Description	Verify system correctly identifies walking action
Test Data	Video of child performing walking action
Test Steps	Navigate to “Start Recording” > Child perform walk > Perform action > Click “Upload”
Expected Result	System correctly identifies action as "Walk", System displays matching percentage, System suggests appropriate difficulty level

Actual result	System identified action as "Walk" with 87% accuracy and suggested "Hard" difficulty level
Status	Pass
Priority	High

Table 13: Test Case – GMS12

2.3.7.13. GMS13

Field	Information
Test ID	GMS13
Test Scenario	Incorrect action performance
Test Description	Verify system correctly identifies when wrong action is performed
Test Data	Child performing standing instead of walking
Test Steps	Random video “Walk” > Click to “Start Recording” > Child perform walk > Perform action > Click “Upload”
Expected Result	System identifies action as "Stand" rather than "Walk", System displays low matching percentage, System suggests appropriate difficulty level
Actual result	System incorrectly identified action as "Walk" with 65% matching
Status	Fail
Priority	High

Table 14: Test Case – GMS13

2.3.7.14. GMS14

Field	Information
Test ID	GMS14
Test Scenario	Level classification - Easy
Test Description	Verify system correctly categorizes low performance to Easy level
Test Data	Performance with matching percentage between 1-40%
Test Steps	System analyzes child's performance > Matching percentage calculated between 1-40%
Expected Result	System suggests "Easy" difficulty level for subsequent activities
Actual result	System suggested "Easy" difficulty level for 35% matching performance
Status	Pass
Priority	High

Table 15: Test Case – GMS14

2.3.7.15. GMS15

Field	Information
-------	-------------

Test ID	GMS15
Test Scenario	Level classification – Medium
Test Description	Verify system correctly categorizes medium performance
Test Data	Performance with matching percentage between 41-79%
Test Steps	System analyzes child's performance > Matching percentage calculated between 41-79%
Expected Result	System suggests "Medium" difficulty level for subsequent activities
Actual result	System suggested "Medium" difficulty level for 65% matching performance
Status	Pass
Priority	High

Table 16: Test Case – GMS15

2.3.7.16. GMS16

Field	Information
Test ID	GMS16
Test Scenario	Level classification - Hard
Test Description	Verify system correctly categorizes high performance to Hard level
Test Data	Performance with matching percentage between 80-100%
Test Steps	System analyzes child's performance > Matching percentage calculated between 80-100%
Expected Result	System suggests "Hard" difficulty level for subsequent activities
Actual result	System suggested "Hard" difficulty level for 92% matching performance
Status	Fail
Priority	High

Table 17: Test Case – GMS16

2.3.7.17. GMS17

Field	Information
Test ID	GMS17
Test Scenario	Activity before random generation
Test Description	Verify system prevents upload before generating random activity
Test Data	N/A
Test Steps	End the process of initial level > Navigate to next level > Click "Upload" without generating random video

Expected Result	System displays notification: "Please generate a random category first"
Actual result	No error message displayed, system attempted to process non-existent video
Status	Fail
Priority	Low

Table 18: Test Case – GMS17

2.3.7.18. GMS18

Field	Information
Test ID	GMS18
Test Scenario	Easy level activity generation
Test Description	Verify appropriate activities are generated for Easy level
Test Data	Child performance categorized as "Easy" level
Test Steps	Navigate to Next Level > Click “Generate Random Video” >
Expected Result	System randomly selects from: Walk, Stand, Catch activities
Actual result	System randomly selected "Stand" activity from appropriate Easy level activities
Status	Pass
Priority	High

Table 19: Test Case – GMS18

2.3.7.19. GMS19

Field	Information
Test ID	GMS19
Test Scenario	Medium level activity generation
Test Description	Verify appropriate activities are generated for Medium level
Test Data	Child performance categorized as "Medium" level
Test Steps	Navigate to Next Level (Medium) > Click “Generate Random Video” >
Expected Result	System randomly selects from: Run, Throw activities
Actual result	System randomly selected "Run" activity from appropriate Medium level activities
Status	Pass
Priority	High

Table 20: Test Case – GMS19

2.3.7.20. GMS20

Field	Information
-------	-------------

Test ID	GMS20
Test Scenario	Hard level activity generation
Test Description	Verify appropriate activities are generated for Hard level
Test Data	Child performance categorized as "Hard" level
Test Steps	Navigate to Next Level (Hard) > Click "Generate Random Video" >
Expected Result	System randomly selects from: Kick a ball, Hand Stand, Dribble activities
Actual result	System selected "Dribble" activity from appropriate Hard level activitie
Status	Pass
Priority	High

Table 21: Test Case – GMS20

2.3.7.21. GMS21

Field	Information
Test ID	GMS21
Test Scenario	Tutorial video delivery
Test Description	Verify system provides appropriate tutorial vide
Test Data	"Walk" activity generated randomly
Test Steps	System generates "Walk" as random activity
Expected Result	System provides video tutorial with appropriate instructions and music
Actual result	System provided appropriate tutorial video but without audio
Status	Partial Pass
Priority	Medium

Table 22: Test Case – GMS21

2.3.7.22. GMS22

Field	Information
Test ID	GMS22
Test Scenario	User account creation
Test Description	Verifying new users can create accounts successfully
Test Data	Valid name, email, password
Test Steps	Navigate to registration page > Complete all required fields > Submit registration form
Expected Result	Receive a pop up
Actual result	Receive a pop up
Status	Pass
Priority	High

Table 23: Test Case – GMS22

2.3.7.23. GMS23

Field	Information
Test ID	GMS23
Test Scenario	Recovery from interrupted upload
Test Description	Verify system can resume interrupted video uploads
Test Data	Large video file, means to interrupt network
Test Steps	Click the “Upload” button > Uploading large video >
Expected Result	System either resumes upload or provides clear error recovery options
Actual result	Upload failed with generic error message; no option to resume where it left off
Status	Fail
Priority	Medium

Table 24: Test Case – GMS23

3. Results & Discussions

3.1 Result

The newly developed basic skills development system was implemented on a large scale to develop the physical skills of Down syndrome. 15 Down syndrome children were randomly chosen for this experimental implementation. Their parents' participation was also valued in this process. The children were given activities for a week to study and report on this.

After the children had passed through the activities of the system, their skill level was assessed and classified. Here, 10 children were classified as "Easy" level, and 3 children as "Medium". Only the remaining 2 were initially classified as "Hard". After this, training activities were carried out under proper guidance for a week.

What is remarkable here is that 70% of the children who were initially at "Easy" levels have been raised to "Medium" level. It was even observed that some of the children who were at the "medium" level even progressed to the "Hard" level. This progress justified the effectiveness of the system and the capabilities of the children in attaining proper skills.

However, there were some challenges, if any, in implementing this process. The environment in which some of the sessions were conducted was full of noise, and the children lost their focus. Additionally, the physical abilities and comfort zones of the selected children varied, and it was a challenge to provide all the children with the same training process.

However, the reaction of the parents was very valuable and favorable. They believed that the system reduced the need to hire professional trainers to enhance the fundamental physical capacities of their kids. They considered the ability to communicate with the children and provide additional instructions, and the ease of movement instructions contained in the system, a very valuable service.

The most important innovative part of this system is that it uses a machine learning and deep learning -based solution to accurately track the movement of children and compute their skill level. The module, which was trained for this purpose, had 90% accuracy. This confirmed that it can track children's activity and provide more accurate and reliable results for children and parents.

This system was in its nascent stage, but today it has become an application that has carved a new way in practical techniques of improving the quality of life of children and providing confidence and security to parents.

3.2 Research Findings

Through this research, the system developed to build fundamental motor skills in Down syndrome was proven to be highly effective. Baseline rated 10 of the 15 children as "Easy", 3 as "Medium", and the rest as "Hard". 70% of the children rated as "Easy" scored a lot of improvement and were reclassified to the "Medium" level during the follow-up. Also, some of the "Medium" children upgraded to the "Hard" level. The gradual improvement can be seen as a major reason for the system's success.

Also, the model that was trained to classify the skill levels of the children and to judge progress had a high accuracy of 90%, which ensured the reliability of the system.

The response from parents who used the system was also very good, stating that they were very happy with the measurable improvement in motor skills of their children and that this method was a very easy way that could even be used at home. Also, because no additional physical trainer was needed by families, their financial burden could be reduced easily, and they could monitor the performance of the child very well.

However, the implementation of the system was not smooth, and the noisy environment during the sessions

caused distraction and reduced participation by some children. Moreover, the vast range of children's personal physical skills and capabilities made it difficult to implement the sessions in a uniform manner to all children.

But even with these challenges, the entire project was successfully implemented. This more or less guarantees that this new system, designed to improve the intellectual potential of Down syndrome kids is useful and of the highest priority to families.

3.3 Discussion

It was the intention of this study to create a sound methodology for improving the fundamental cognitive abilities of children with Down Syndrome. The tailor-made system that was created for this was tried out on 15 children and their parents. At the outset, 10 of the children were "Easy", 3 were "Medium", and the other children were "Hard".

It was intriguing that throughout the continuous testing period while using the system, 70% of the children who were "Easy" progressed to "Medium." Furthermore, a few of the children who were "Medium" progressed to "Hard" as well. This progress is good evidence of the effectiveness of the system and that it has had a direct impact on the development of the children's skills.

The AI model, once trained, was found to be very accurate with 90% accuracy and could analyze the performance of the child more accurately and provide feedback according to ability. This kind of reliability was a great help for parents as they did not have to use an outside physical trainer to enhance the motor skills of their children.

The response of the parents was extremely positive, indicating that the system was a simple, helpful and handy way to develop their children's abilities, and that they were able to provide additional support to their children when needed.

There were, nevertheless, some problems in the system implementation. Some sessions were conducted where there was a lot of noise, and this had the effect of negatively impacting children's participation and attention. Secondly, the speed of adjustment and response to activities varied from session to session due to the differences in children's physical abilities.

Parents simultaneously reported that the inbuilt chaos actions in this system were simple and clear for the children to replicate, and it was very easy to provide them with additional practice at home.

This research shows that artificial intelligence and technology can be employed in developing the chaos capacity of children with Down syndrome in a complex yet effective way.

4. Conclusion

This research main objective was to create a simple, effective and sustainable method of developing the basic motor skills of children with Down syndrome. For this purpose, the gross motor skill development system created was tested with 15 children and their parents.

By implementing the training programs in the system in a systematic manner, very good results were seen. 70% of the children who were initially categorized as "difficult" shifted to the "medium" level, and some children in the "medium" level also shifted to the "high" level. This shows that the system designed successfully impacted the children's motor skill development.

The performance analysis model of the children using artificial intelligence (AI) technology was also extremely successful. This model, which provided 90% accuracy, was useful in determining the advanced abilities of the children. Accordingly, it was possible to provide proper feedback according to the level of development of the child.

Feedback from parents also made the value of the system understandable. They pointed out that the system had enabled them to promote their children's motor skills more easily, affordably and safely. The fact that parents themselves were able to learn the activities and assist the children more where needed was another positive factor.

But challenges were also considered at the time of implementation. There were some situations where some of the sessions took place in noisy locations, and children would become distracted. Additionally, the physical ability variation of the children indicated some constraints to adhering to a standardized training scheme.

But despite all these challenges, the research showed that it was possible to make this system an accessible, reliable and feasible solution for the entire populace. There were future opportunities to make this project even more productive and useful by continuing to develop the technology, expanding the training programs and adding more methods to the system.

Lastly, this research was an important contribution to the improvement of the quality of life of children with Down syndrome. Arguably, the system of development of gross motor skills has brought about a new perspective to the lives of children and parents.

4.1. Future Workings

A personalized activity recommendation system can be developed that will automatically provide recommendations based on the child's unique abilities, age, and developmental needs. This is achieved by developing AI technology and reviewing the child's past data. Second, new activities can be added to build complex coordination, balance, and speed with basic movement skills. Third, a system of immediate feedback should be employed that possesses the ability to provide accurate instructions and feedback as the child is executing. This will enable them to master the activities accurately and in depth. Fourth, deeper models of higher order can be deployed that will better analyze the child's verticality and patterns of movement. Fifth, there should be a established system of reporting that has the capability of providing daily progress reports to parents and educators. This will be better placed to measure the child's future activities and long-term development. Future work plans could include providing this service to additional children via the Internet, developing fitting software in due course, and including visual and audio-visual instructions for ease of use.

5. References

- [1] Scottish Down's Syndrome Association, *What is Down Syndrome?*, Scottish Down's Syndrome Association, 2001.
- [2] Erin E Wentz, Julia Looper, Kristi S Menear, Dipika Rohadia, Nora Shields, Promoting Participation in Physical Activity in Children and Adolescents With Down Syndrome, *Physical Therapy*, Volume 101, Issue 5, May 2021, pزاب032
- [3] M. Horvat, R. Croce, and A. Fallaize, "Information processing and motor control in Down syndrome," *Journal of Down Syndrome & Chromosome Abnormalities*, vol. 2, no. 1, p. 107, 2016
- [4] R. Cui, A. Zhu, G. Hua, H. Yin, and H. Liu, "Multisource learning for skeleton-based action recognition using deep LSTM and CNN," *Journal of Electronic Imaging*, vol. 27, no. 4, pp. 043050-043050, 2018

- [5] L. C. Dapp, V. Gashaj, and C. M. Roebbers, "Physical activity and motor skills in children: A differentiated approach," *Psychology of Sport and Exercise*, vol. 54, pp. 101916, 2021
- [6] R. Malak, M. Kotwicka, A. Krawczyk-Wasielewska, E. Mojs, and W. Szamborski, "Motor skills, cognitive development and balance functions of children with Down syndrome," *Annals of Agricultural and Environmental Medicine*, vol. 20, no. 4, 2013
- [7] H. M. C. H. Herath, N. A. S. B. Nissanka, I. A. Illankoon, R. P. Madusanka, J. Krishara, and W. Tissera, "EduPlanner–Best Teaching Method for Students with Down Syndrome," in *2023 5th International Conference on Advancements in Computing (ICAC)*, Dec. 2023, pp. 304-309. IEEE
- [8] N. M. Safri and R. T. Y. Sheng, "Surface Electromyographic Signals of Special Needs Children during Fine Motor Task," in *2019 International Biomedical Instrumentation and Technology Conference (IBITeC)*, vol. 1, Oct. 2019, pp. 17-20. IEEE
- [9] W. M. Shalash, S. AlTamimi, E. Abdu, and A. Barom, "No Limit: a down syndrome children educational game," in *2018 IEEE Games, Entertainment, Media Conference (GEM)*, Aug. 2018, pp. 352-358. IEEE.
- [10] A. Aydogan and M. S. Demirok, "The effectiveness of computer aided video modeling in teaching basic basketball movements to individuals with Down syndrome," *Children*, vol. 10, no. 1, p. 153, 2023
- [11] S. Chun, S. Jang, J. Y. Kim, C. Ko, J. Lee, J. Hong, and Y. R. Park, "Comprehensive Assessment and Early Prediction of Gross Motor Performance in Toddlers With Graph Convolutional Networks–Based Deep Learning: Development and Validation Study," *JMIR Formative Research*, vol. 8, e51996, 2024
- [12] M. Ahmadi, M. O’Neil, M. Fragala-Pinkham, N. Lennon, and S. Trost, "Machine learning algorithms for activity recognition in ambulant children and adolescents with cerebral palsy," *Journal of Neuroengineering and Rehabilitation*, vol. 15, pp. 1-9, 2018
- [13] H. H. Kim, J. Y. Kim, B. K. Jang, J. H. Lee, J. H. Kim, D. H. Lee, ... and Y. R. Park, "Multiview child motor development dataset for AI-driven assessment of child development," *GigaScience*, vol. 12, giad039, 2023
- [14] L. D. C. de Menezes, T. Massetti, F. R. Oliveira, L. C. de Abreu, S. R. P. Malheiros, I. L. Trevizan, ... and C. B. de Mello Monteiro, "Motor learning and virtual reality in Down syndrome; a literature review," *International Archives of Medicine*, vol. 8, 2015.
- [15] Y. Z. Yang, J. A. Tsai, Y. L. Yu, M. H. J. Ko, H. Y. Chiou, T. W. Pai, and H. J. Chen, "Automatic Movement Recognition for Evaluating the Gross Motor Development of Infants," *Children*, vol. 12, no. 3, p. 310, 2025.
- [16] Prasad, K.V., Kishore, P.V.V., Rao, O.S., 2020. Convolutional Long

Short-Term Memory Hybrid Networks for Skeletal Based Human Action Recognition.
International Journal of Innovative Technology and

Exploring Engineering (IJITEE), 9(3), pp.423-430.

[17] 12. Assessment of developmental learning and behavioural problems in children...

Trevor S Parry

Medical Journal of Australia; Jul 4, 2005; 183, 1; ProQuest Medical Library